GNG2101

Design Project Progress Update

"CTRL Sync"
GROUP #3.1

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Table of Contents

List of Figures iv

List of Tables v

List of Acronyms and Glossary vi

1. Introduction 1

    1.1 Assumptions 2

2. Sustainability Report and DFX 3

    2.1 Sustainability Report 3

    2.2 LCA Report 8

    2.3 DFX …….....17

    2.4 AI Appendix 20

3. Problem Definition, Concept Development, and Project Plan 23

    3.1 Problem Definition 23

    3.2 Concept Development 32

    3.3 Project Plan 40

    3.4 AI Appendix 41

4. Detailed Design and BOM 43

    4.1 Problem Definition .....

    4.2 Concept Development .....

    4.3 Project Plan .....

    4.4 AI Appendix .....

4.5 Problem Definition .....

4.6 Concept Development .....

List of Figures

Figure 1: Client's Natural One-Handed Grip on Standard Controller 2

Figure 2: Goal and Scope Definition 8

Figure 3: System Boundary 10

Figure 4: Inventory Analysis 13

Figure 5: Impact Analysis 15

Figure 6: Interpretation and Design Improvements 16

Figure 7: Example of AI Use for Understanding Design for X 21

Figure 8: Example of AI Use for Understanding Design for X 22

Figure 9: Example of AI Use for Estimating ABS CO_2 Emission Factor in the LCA 22

Figure 10: Xbox Adaptive Controller Interface Layout (Microsoft) 29

Figure 11: Picture of the Attachment from the Reference Video) 30

Figure 12: Conceptual Sketch of Dual Lever Arm Mechanism with Spring Return 37

Figure 13: Front View of the CAD Model 38

Figure 14: Angled View of the CAD Model 38

Figure 15: Updated Gantt Chart 40

Figure 16: Project Task List Showing Deliverables, Milestones, and Deadlines 40

Figure 17: Example of AI-Assisted Figure Title Generation 41

Figure 18: Example of AI Use for Pros and Cons Analysis During Concept Selection 42

List of Tables

Table 1. Acronyms vi

Table 2. Glossary vi

Table 3. Triple bottom line table 6

Table 4. Elements inside the boundary 9

Table 5. Elements outside the boundary 10

Table 6. Material inputs 13

Table 7. Emission calculation 14

Table 8. Measurable Metrics Derived from Client Needs 25

Table 9. Target Specifications for Add-On Attachment 31

List of Acronyms and Glossary

Table 1. Acronyms

Acronym	Definition
ABS	Acrylonitrile Butadiene Styrene
BOM	Bill of Materials
DFX	Design for X
FPS	First Person Shooter
LCA	Life Cycle Analysis
PLA	Polylactic Acid

Table 2. Glossary

Term	Acronym	Definition
Add-on Attachment		A device that attaches to an existing product to add functionality.
Adaptive Device		A device designed to assist users with physical impairments.
Bill of Materials	BOM	A detailed list of all materials, components, and quantities required to manufacture a product.
Cradle-to-Grave		An LCA approach covering a product's entire life cycle from creation to disposal.
Design for X	DFX	A design approach that optimizes a product for specific objectives.
First Person Shooter	FPS	A game genre played from the character's point of view that involves aiming and shooting.
Life Cycle Analysis	LCA	A method for evaluating environmental impacts across a product's life cycle.
Modular Design		A design using replaceable components to improve repairability.

1 Introduction

One of the earliest video games ever made was developed in 1958 by American physicist William Higinbotham. Since then, the gaming industry has experienced exponential growth. In today's age, according to Fabio Duarte, over "3.32 billion people are actively playing" [1] in the year 2026, with all types of different demographics. However, despite this rapid growth and technological advancement, many gaming systems are still designed with the assumption that users have full use of both hands. Modern controllers require coordinated two-handed operation to access all buttons, triggers, and joysticks simultaneously. For individuals with limited or no use of one hand, this creates significant barriers. As a result, accessibility remains a crucial yet often overlooked aspect of controller design.

This design project focuses on developing an adaptive add-on attachment for a Nintendo Switch Pro Controller that improves accessibility for users who have limited or no use of one hand. The objective is to create a modular, eco-friendly, and cost-effective solution that allows the users to operate essential controls with a single hand without permanently removing that natural feeling some one-handed users have already developed. Essentially, the design aims to maintain or maybe even enhance gameplay performance while improving comfort, control, and independence for the user.

This document is organized to clearly present the progression of our design project in a logical and structured manner. focuses on sustainability considerations, including an LCA that evaluates the environmental impacts of our material and design choices. DFX principles are then discussed to justify key decisions related to accessibility, assembly, durability, sustainability, and cost. This is then accompanied by problem definition which gives details of the needs of the client, constraints, and other background information. The concept development process is then described in the report, which also explains the process of generating and evaluating solutions. A project plan is included to demonstrate how tasks were organized and managed throughout the design process. In the detailed design part, the final concept and most important technical choices are given. A BOM is also included to state the elements that are needed in the implementation. Overall, the purpose of this is to document and justify our engineering decisions while

demonstrating how our proposed solution effectively addresses the client's accessibility needs through a systematic and evidence-based design process.

1.1 Assumptions

The first assumption in this project is that a significant number of one-handed gamers get used to playing with a controller in the same manner that our client is used to playing with creativity in positioning and alternative control techniques to compensate for limited mobility. We made this assumption because our initial client meeting provided the clearest insight into real-world usage, and our client's experience served as a practical and observable starting point for design decisions. The method provided by him served as a base on which we could base our design instead of drawing generalizations that may not be supported. An image of his grip of a Nintendo Switch Pro Controller will be presented below. Another assumption that was made was that users would be willing to use an attachment-based solution rather than buying an entirely redesigned controller as this approach is more cost-effective and preserves familiarity with equipment they already own. This was presumed to make the design accessible, realistic as well as in accordance with the preferences of the client. Other than these user-based considerations, there were no other assumptions that were made when developing this project.

Figure 1: Client's Natural One-Handed Grip on Standard Controller

2 Sustainability Report and DFX

2.1 Sustainability Report

2.1.1 Economic

Positive Impact:

1. The one-handed controller is designed for a specific accessibility market that is often overlooked by gaming companies. By creating a one-handed controller, this helps a company/group stand out from competitors that mostly focus on standard controllers. Since the product addresses a significant accessibility need, customers may perceive strong value in it, even if it is priced similarly to standard controllers. As awareness of accessible gaming continues to grow in the current day, demand for this type of product is likely to increase, leading to more stable sales over time.
2. There is strong potential for long-term revenue as accessibility becomes a bigger priority in the gaming industry. Although it is still often overlooked, an increasing number of consumers and companies are beginning to value inclusive design. This could create opportunities to partner with gaming studios, healthcare organizations, or accessibility groups. These partnerships could help build trust in the product while reducing the need for heavy marketing and advertising.
3. By focusing on a specialized product, a company/group avoids direct competition with large mass-market controller brands. This means there is less pressure to compete mainly with prices. Instead, the product stands out because of what it offers and the positive impact it can have. This approach can help build customer loyalty and strengthen a company's/group's overall brand reputation.

Negative Impact:

1. Because the controller is a specialized product, it will likely be produced in smaller quantities than standard controllers. Lower production volumes increase the cost per unit and limit economies of scale. This can make it more difficult to keep the product

affordable while still making a profit. As a result, there is greater financial risk, especially during the early stages of production.

2. The use of custom components and molds adds complexity to the manufacturing process. These requirements lead to higher setups and tooling costs. Extra testing is also needed to make sure the controller meets durability and accessibility standards. Together, these factors increase the amount of money required upfront.
3. Demand for the controller may be hard to predict because it targets a niche market. Sales could vary depending on factors like public awareness, funding programs, or changing consumer trends. This uncertainty makes it more difficult to estimate future revenue. Uneven sales can also create challenges when planning inventory and managing cash flow.

2.1.2 Environmental

Positive Impact:

1. The controller can be designed to last longer, which reduces how often it needs to be replaced. This helps lower material usage and reduce waste over time. Overall, longer-lasting products tend to have a smaller environmental impact.
2. A modular design would allow certain parts, such as buttons or joysticks, to be replaced instead of replacing the entire controller. This encourages users to repair the product rather than discard it. Repairability helps reduce electronic waste and extends the usable life of the controller. It also promotes more sustainable consumer habits.
3. The controller can be designed to work with existing Nintendo Switch hardware and accessories. This removes the need for extra adapters or converters. Using fewer accessories helps reduce material use and packaging waste. Overall, this makes the system more resource efficient.

Negative Impact:

1. The controller requires plastics and maybe some metal components, which all have environmental impacts during extraction and manufacturing. These processes contribute to greenhouse gas emissions and the use of limited resources. Plastics are especially concerning because they take a long time to break down. This impact is difficult to fully avoid with electronic products.
2. Creating an add-on attachment for an existing controller requires extra materials that would not be needed otherwise. Even if it does not use batteries, the attachment still relies on plastics, fasteners, and possibly wiring or mechanical parts. This increases the total amount of material used per user, since both the original controller and the add-on must be manufactured. As a result, overall resource use may increase instead of decrease.
3. Domestic and global manufacturing and distribution increase the carbon footprint of the product. Shipping components and finished products over long distances consumes fossil fuels. This impact grows as production scales up. Transportation emissions are a significant part of the product's life cycle.

2.1.3 Social

Positive Impact:

1. The controller improves accessibility for people with limited use of one hand, allowing them to play video games on their own. This helps promote inclusivity and gives more people equal access to entertainment. Users who might otherwise be left out are able to fully take part in gaming. This can have a positive effect on their overall quality of life.
2. Greater access to gaming also supports social interaction and connection. Multiplayer games allow users to play with friends and family, which can help reduce feelings of isolation, especially for individuals with disabilities. Gaming can also be a way to relax and relieve stress, which may benefit mental health.
3. The product also helps raise awareness about the importance of inclusive and accessible design. It encourages engineers and companies to think more carefully about different

user needs. This mindset can influence future products in the gaming industry. Over time, it can help create a more inclusive society overall.

Negative Impact:

1. The add-on may not work equally well for all one-handed users. Differences in hand size, grip strength, or range of movement can affect how easy it is to use. A one-size-fits-all design may not meet everyone's needs. As a result, some users may still feel excluded.
2. Limited marketing or distribution could make it difficult for the controller to reach the people it is meant to help. Many users who would benefit from accessible gaming options may not even know these types of controllers exist. Without targeted outreach through accessibility groups, healthcare providers, or gaming communities, awareness may stay low. This can reduce adoption and limit the product's overall social impact. Clear communication and effective distribution are important to ensure the controller reaches the users it is designed for.
3. The potential of the higher cost of a specialized controller could make it harder for some users to access. People who would benefit the most may not be able to afford it. This creates a financial barrier to accessibility. As a result, not everyone may be able to experience the social benefits of the product.

2.1.4 Table for the Report

Table 3. Triple bottom line table

Triple Bottom Line	Positive Impact	Negative Impact
Economic	Targets a frequently neglected accessibility market, helping the product stand out as demand grows.	Low production volumes increase per-unit costs and financial risk.
	Growing emphasis on inclusive design could lead to long-term revenue potential and partnership opportunities.	Custom components and testing raise manufacturing and upfront costs.
	Specialized product reduces price competition and helps build customer loyalty and brand reputation.	Niche market demand is uncertain, complicating revenue and inventory planning.
Environmental	Designing for longer product lifespan reduces material use and waste.	Plastic and metal components contribute to resource depletion and emissions.
	A modular and repairable design extends usability and lowers waste.	Add-on attachments require extra materials, increasing overall resource use.
	Compatibility with existing hardware reduces additional materials and packaging.	Manufacturing and long-distance transportation increase the product's carbon footprint.
Social	Improves gaming accessibility for one-handed users, promoting inclusion and quality of life.	One-size-fits-all design may not meet the needs of all one-handed users.
	Increased access to gaming supports social connection, stress relief, and mental well-being.	Limited marketing or distribution may reduce awareness and adoption among target users.
	Raises awareness of inclusive design and encourages more accessible future products for anything in general (not just gaming).	Higher costs may create financial barriers, limiting accessibility for some users.

2.2 LCA report

2.2.1 Goal and Scope Definition

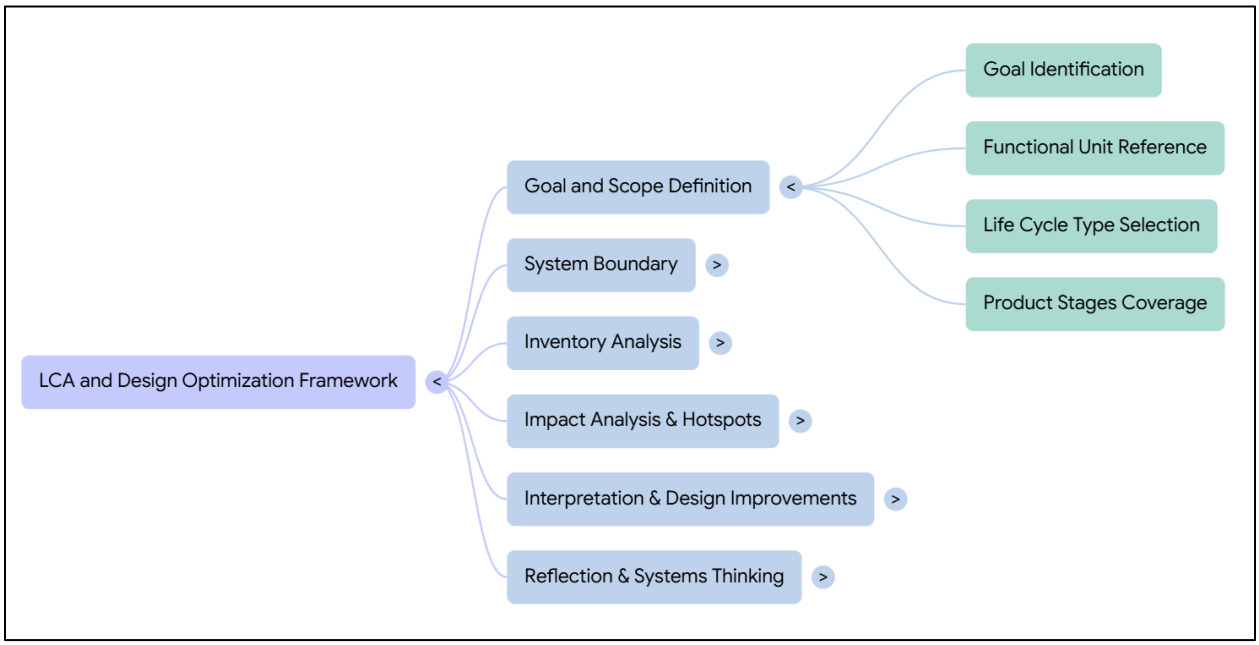


Figure 2: Goal and Scope Definition

2.2.1.1 Goal

The purpose of our life cycle analysis is to compare the environmental impacts of design and material choices for an adaptive Nintendo switch add on, to identify places where we can optimize environmental impact while balancing the triple bottom line framework for ideal sustainability.

2.2.1.2 Functional Unit (This is your reference measure)

The functional unit is the standard amount of product use against which all environmental impacts are measured. It ensures fair comparison.

To ensure a fair and consistent comparison between design options our functional unit will be: One device used daily for 25 years (approximately 9125 days (about 25 years)

Why did we choose this: Because after getting to know our client we learned that he has a strong passion for gaming and finds himself with a lot of free time. For that reason, it would be appropriate to only consider design options that are meant to last a large portion of a lifetime. Choosing a functional unit of 25-year assuming daily use, allows the environmental impact to be evaluated on a lifetime basis rather than per individual use, providing a more meaningful and useful evaluation.

2.2.1.3 Life Cycle Type

We have chosen to use a cradle-to-grave life cycle analysis, as it best reflects the long-term environmental impact of a 25-year assistive device. Our product stands out due to its long lifespan, which ultimately results in a lower environmental impact per use. This is not only beneficial from a product marketing standpoint but also encourages consideration of materials with appropriate end-of-life options, aligning the design with sustainability goals.

2.2.1.4 Product Stages Covered

The following stages will be included in our LCA?

1. **Raw material Extraction & Processing**

   Include:

   - Extraction of base materials
   - Processing and refining of materials into usable forms (pellets, sheets, filaments)

2. **Manufacturing**

   Include:

   - Fabrication of individual parts
   - Estimated Energy and material input

3. **End of Life**

   Include:

   - Disposal (Landfill, recycling)
   - Material recovery

2.2.2 System Boundary

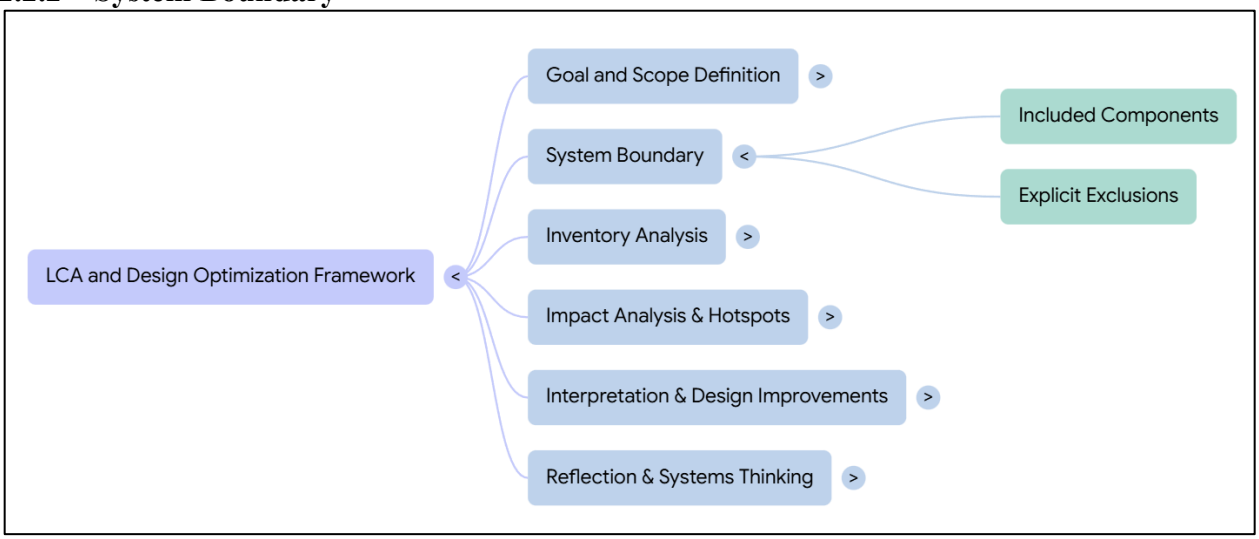


Figure 3: System Boundary

2.2.2.1 What's Inside the Boundary (Included in the LCA)

Table 4. Elements inside the boundary

What's Included	Why
Raw material extraction and processing - Extraction of the plastics - Extraction of ecofriendly screws and fasteners)	With our design potentially being primarily made from plastic filament, it is important for us to prioritize using 3D printing filaments made from green process with recyclable characteristics.
End-of-life treatment (disposal/recycling) - Estimated Energy use for disposal - Estimated Carbon emissions	Considering end-of-life treatment is essential when designing an environmentally friendly product. Therefore, it is crucial to evaluate the end-of-life options of the materials selected for our design.

2.2.2.2 What's Outside the Boundary (Explicitly Excluded)

Table 5. Elements outside the boundary

What's Excluded	Why
Electricity consumption of the gaming console	Console energy use is unrelated to the assistive device and would not vary between design options.

Transportation Cost - Raw material transportation - Product shipping - Employee commuting	Because transportation can be hard to measure and isn't something we can easily optimize for.
Maintenance - Expected replacement components	Maintenance was excluded because the device is designed for passive mechanical use and durability. Component replacement is not expected over its service life.
Component manufacturing (COULD BE SUBJECT TO CHANGE) - Energy used during the 3D printing process.	Our manufacturing flexibility is limited, as we are restricted to using the school's 3D printers and, if incorporated, the design and manufacturing of the leg straps are unclear and out of our control.
Capital Goods	Excluded for simplicity and lack of relevancy
Processes and environmental impact of Screws fasteners, Magnets and springs	Screws fasteners and magnets are small and they're limited alternative choices
Use Phase	There are no sustainability-related factors that need to be evaluated during this phase for our specific product.

2.2.3 Inventory Analysis

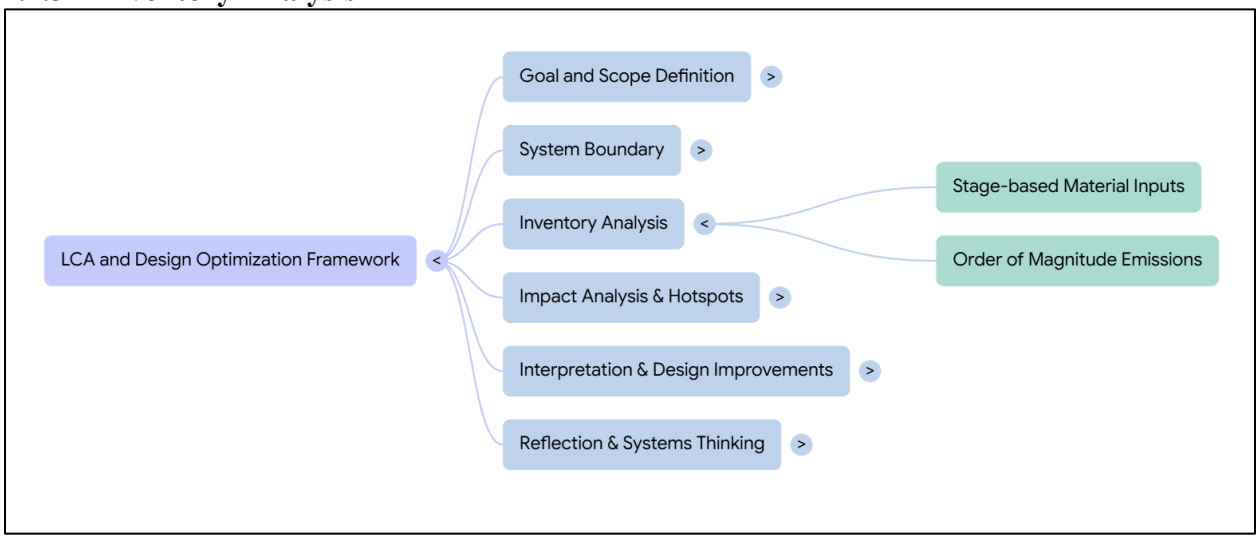


Figure 4: Inventory Analysis

Material Inputs by Stage

Raw Material Extraction Stage

Table 6. Material inputs

Product Stage	Material/Energy Input	Quantity	Notes
Raw material extraction	ABS	Approx. 20-30 grams	ABS will be the main material used in our design
ABS Manufacturing	Electricity use	0.3 kW ≈ 18–28 grams of CO_2	Printing 25 grams worth of parts will typically take around 2 hours on a Bambu P1S (our printer of choice) Average 3-hour print ≈ 300 W Energy = 0.3 x 2 = 0.6 kWh Ontario Estimated CO2 emission 0.6 kWh x 30 g/ kWh = 18 g CO_2 0.7 kWh x 40 g/ kWh = 28g CO_2

End of life	Recycling or incinerator of ABS plastic	Approx. 20-30 grams	ABS is not commonly accepted by Municipalities recycling

Rough Emissions Calculation (Order of Magnitude)

Table 7. Emission calculation

Life Cycle Stage	Calculation	CO_2e Output
Raw Material Extraction (ABS production)	0.025 kg × 2.7 kg CO_2/kg	0.0675 kg = 67.5 g CO_2
Manufacturing – Electricity (low estimate)	0.6 kWh × 30 g/kWh	**18 g CO_2**
Manufacturing – Electricity (high estimate)	0.7 kWh × 40 g/kWh	28 g CO_2
End of Life – Landfill (no energy recovery)	Negligible direct emissions (plastic stable)	~0 g
End of Life – Incineration (if burned)	0.025 kg × ~3 kg CO_2/kg plastic	~75 g CO_2
	TOTAL per functional unit	If landfilled: 85–95-gram CO_2 If Incinerated: 160-170 g CO_2

2.2.4 Impact Analysis - Identifying Hotspots

Hotspot: The life-cycle stage where the largest environmental impact occurs.

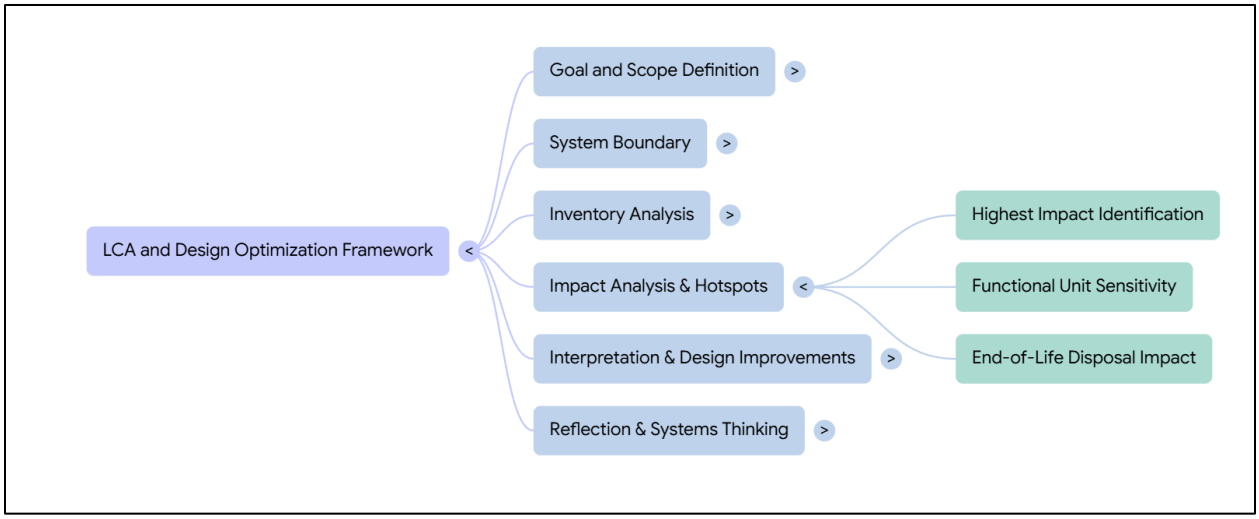


Figure 5: Impact Analysis

Question 1: Where is the biggest impact?

Rank the stages by their contribution to total CO_2e:

Raw Material extraction contributes the least to the CO2 output, since no actual materials are being extracted and the majority of the materials at hand are being sourced in a pre-made state. The manufacturing process (3D printing) may contribute slightly negatively to the C02 output, due to its use of electricity and release of steam, but overall, it's quite minimal.

Question 2: What if we change the functional unit?

If we change the functional unit, in the case of the materials being extracted, the C02 output would increase if the materials required actual machinery to be sourced. This is unlikely in our case, but would increase the output, nonetheless.

Question 3: What if we change the end-of-life disposal?

This question can be answered with greater accuracy, once we know the details of the actual materials being used.

2.2.5 Interpretation and Design Improvements

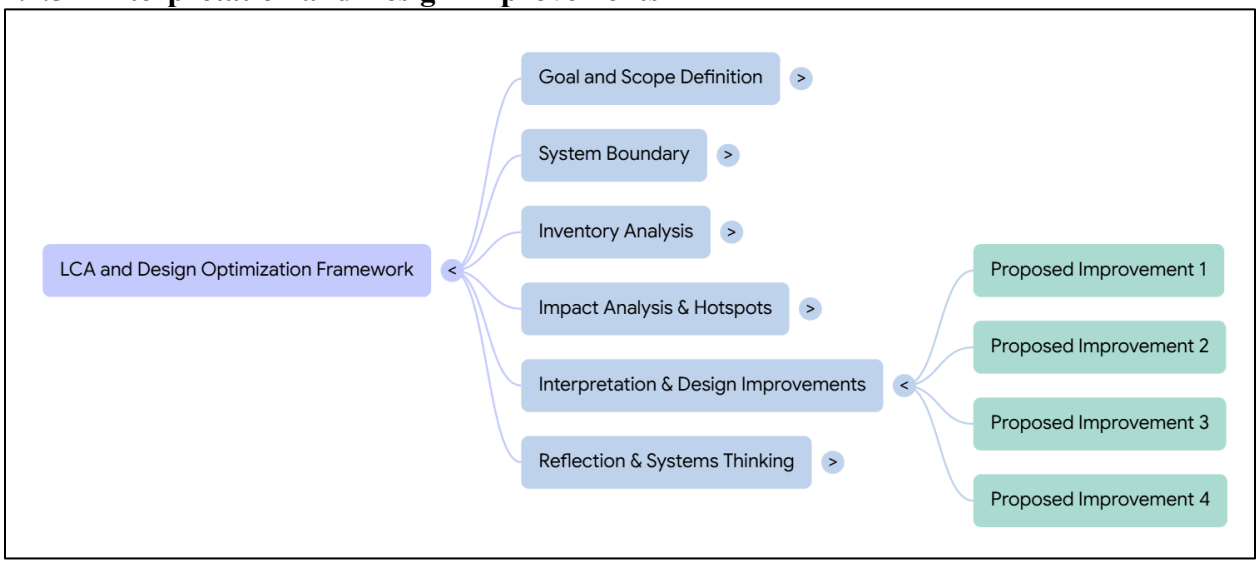


Figure 6: Interpretation and Design Improvements

Improvement #1: Optimize Design to Reduce Material Mass

a) The component would be redesigned to make the walls thinner and have internal lattice structures so as to consume less material.

b) The change influences the manufacturing (3D printing) stage as well as the raw material stage.

c) It is important since reduced quantity of material will result in embodied emissions being less and will also promote a reduced printing time which will consequently lower the consumption of electricity.

d) 20% reduction in material mass would result in a total reduction of emissions of about 15- 25 g CO_2 per unit.

Improvement #2: Optimize Design to Reduce Material Mass

a) The design would be made in such a way that it is easier to disassemble the product hence the finished product can undergo recycling instead of landfill or incineration.

b) This transformation has impact on the stage of end-of-life on a life cycle.

c) This is important since during incineration, there is a substantial CO2 increase over landfill disposal as has been estimated.

d) Recycling the product would prevent no less than about 75g CO2 per unit going to waste instead of undergoing incineration.

2.3 DFX

2.3.1 Five Most Important Factors

Based on our meeting with the client, the five most important Design for X factors are:

1) Design for accessibility
2) Design for assembly
3) Design for reliability and durability
4) Design for sustainability
5) Design for cost and affordability

2.3.2 Justification for Each Choice

1. Design for Accessibility

Goal: Allow the user to operate both sides of a standard controller using one hand.

How we measure it:

- Can the user reach all controls from one side?
- Can the device accurately transfer button and joystick motions?
- Can the user play without external help?

Limits:

- The attachment must work with common controllers.
- It must not block the user's hand or view.
- The mechanism must move smoothly without high force.

Design choices:

- Mechanical arms that press buttons on other side of controller.
- Adjustable link lengths to match different controllers,
- Swivel joints for smooth movement.

Why it matters: The device directly replaces the missing function of the left hand, giving the user independence while playing.

2. Design for Assembly

Goal: Make the attachment easy to install, remove, and adjust.

How we measure it:

- Have several people try to set it up and time each person, then take the average of that time and add a minute. This will be the approximate time needed to set the device.
- Number of parts.
- If it can be done with one hand.

Limits:

- Must attach in a few minutes (the time mentioned above).
- Should not require tools.
- Parts must be large and easy to handle.

Design choices:

- Clip-on mounting system.
- Snap fit joints.
- Color coded adjustment points.

Why it matters: Due to the users' limitations, they should easily be able to set the device up, without needing any help.

3. Design for Reliability & Durability

Goal: Ensure the device functions the exact same over long periods of use (about 3+ hours).

How we measure it:

- Number of movement cycles.
- Resistance to wear.
- Performance after being dropped.

Limits:

- Must survive daily game use.
- Must not loosen over time.
- Must resisting bending and breaking.

Design choices:

- Reinforces joints.
- Flexible but strong plastic.
- Metal pins in high-stress areas.

Why it matters: The arms are constantly moving and are essential, so it must not fail or wear over time.

4. Design for Sustainability

Goal: Reduce environmental impact.

How we measure it:

- Amount of recyclable material.
- Waste from manufacturing.
- Packaging size.

Limits:

- Use recyclable plastics.
- Minimal packaging.
- Modular parts for repair.

Design choices:

- Recycled ABS or PLA.
- Replaceable arms.
- Cardboard packaging.

Why it matters: Designing with sustainability in mind helps reduce waste and unnecessary material use.

5. Design for Cost

Goal: Keep the product affordable.

How we check it:

- Estimated build cost.
- Expected selling price.

Limits:

- Must not exceed 50% of the price of a controller.
- Simple materials

Design choices:

- Design the arms to be made from one molded part instead of multiple parts.
- Design so it can be 3D printed for small batches before mass production.

Why it matters: Keeping the design simple and using common parts helps reduce manufacturing costs, making the device more affordable for users.

2.4 AI Appendix

AI was used in our project in a limited way, mainly to help us better understand certain topics and to fix small grammar or wording issues in the report. It was also helpful for getting a clearer idea of how different sections of the deliverable fit together and what was expected in each part. In some cases, AI was used to help make our writing more condensed while keeping the information clear and concise.

In addition, AI helped us connect concepts learned in class to our design project, especially for topics like sustainability, Design for X, and problem definition. It supported our learning and understanding, but all design decisions and final written content were created and reviewed by the group. Screenshots showing some examples of how AI was used are included below.

AI was also used to help estimate the average values of material mass and energy consumption of some of the components to be part of the LCA. This proved very useful, especially in instances where one might find it hard to find credible information online. The professor stated that it is acceptable to use AI in situations when such information is difficult to access and under the condition of responsible and open use. Amendments to the generated estimates were made depending on the necessity and were validated to comply with concepts of the course, and the ultimate assumptions and conclusions were reached by the group.

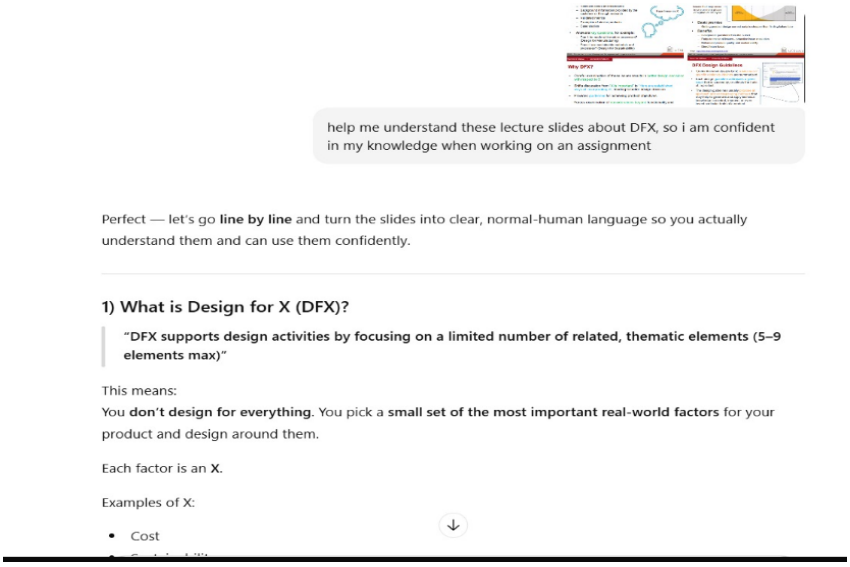
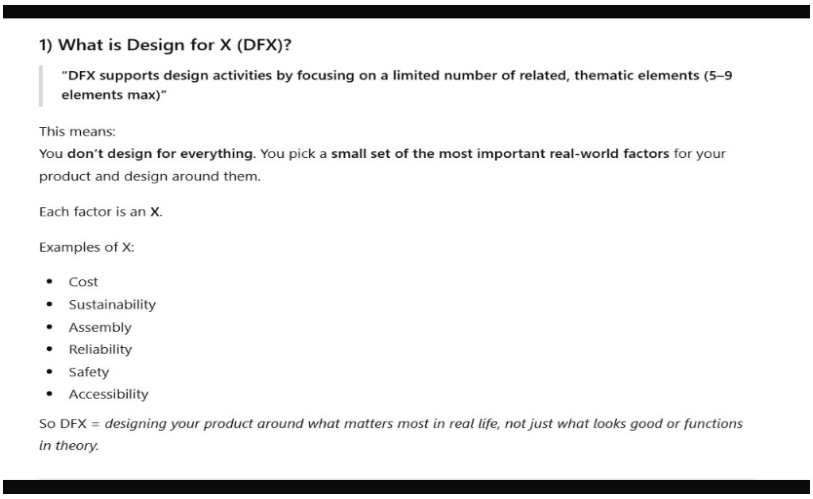


Figure 7: Example of AI Prompts Used for Concept Understanding



1) What is Design for X (DFX)?

> **"DFX supports design activities by focusing on a limited number of related, thematic elements (5–9 elements max)"**

This means:
You **don't design for everything**. You pick a **small set of the most important real-world factors** for your product and design around them.

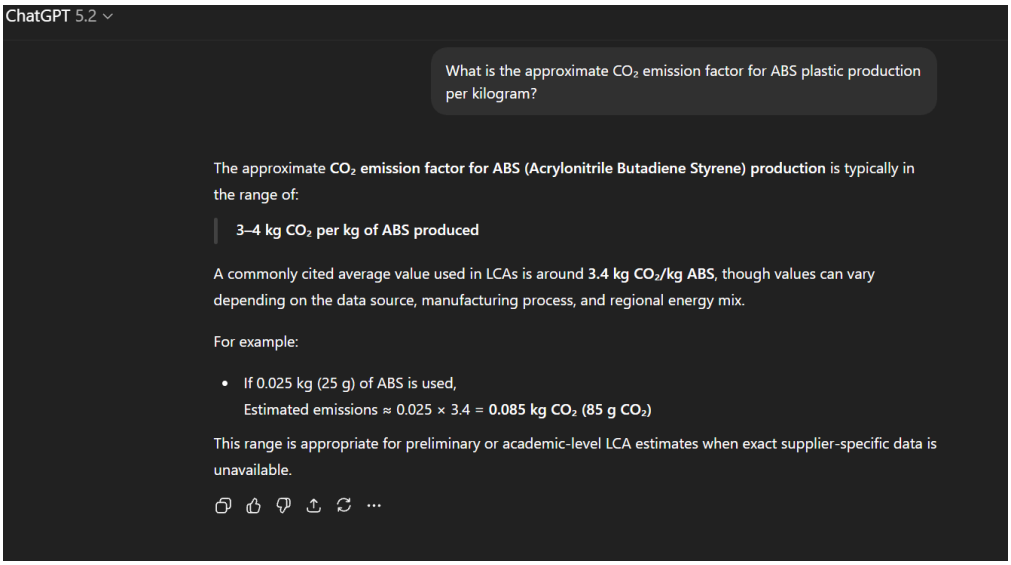
Each factor is an **X**.

Examples of X:

- Cost
- Sustainability
- Assembly
- Reliability
- Safety
- Accessibility

So DFX = *designing your product around what matters most in real life, not just what looks good or functions in theory.*

Figure 8: Example of AI Use for Understanding Design for X



ChatGPT 5.2

What is the approximate CO_2 emission factor for ABS plastic production per kilogram?

The approximate **CO_2 emission factor for ABS (Acrylonitrile Butadiene Styrene) production** is typically in the range of:

> **3–4 kg CO_2 per kg of ABS produced**

A commonly cited average value used in LCAs is around **3.4 kg CO_2/kg ABS**, though values can vary depending on the data source, manufacturing process, and regional energy mix.

For example:

- If 0.025 kg (25 g) of ABS is used,
  Estimated emissions ≈ 0.025 × 3.4 = **0.085 kg CO_2 (85 g CO_2)**

This range is appropriate for preliminary or academic-level LCA estimates when exact supplier-specific data is unavailable.

Figure 9: Example of AI Use for Estimating ABS CO_2 Emission Factor in the LCA

3 Problem Definition, Concept Development, and Project Plan

3.1 Problem definition

3.1.1 Client Needs, Problems, and Known / Unknown Information

After meeting with the client and conducting initial research, we identified that the client has limited use of one hand and would like to independently play Nintendo Switch games that are typically designed for two-handed use, as well as make it easier to continue playing the games he already enjoys, using a controller that he already owns. As a group, we have

<u>Prioritized Client Needs / Problems:</u>

High Priority,

1) Add-on must be compatible and fit with the controller he uses.
2) Ability to access all essential buttons and joysticks on a standard Nintendo Switch controller using only one hand (Including the trigger buttons that are on top of the controller).
3) Device must allow independent use, without assistance from another person.
4) Attachment must be comfortable and not cause strain during long gaming sessions.
5) The device must be reliable and durable for repeated daily use.

Medium Priority,

6) Easy to install and remove.
7) The design should avoid blocking the screen or controller grips and should not restrict natural hand movement.
8) The attachment should be easy to learn and should not require extensive practice before use.

Low Priority,

9) The attachment should not produce excessive noise during use, but minor noise is acceptable.
10) The device should be easy to transport, but portability is not a primary concern.
11) The attachment should have a simple and clean appearance, but aesthetics is less important than functionality.

<u>Known Information:</u>

- The client owns both the Nintendo Switch controller Joy-Cons and the Pro Controller version.
- The main problem is pressing bumpers button and holding the both triggers.
- The client enjoys gaming for long periods of time.
- The client loves playing role-playing games as it is not hard to play.
- The client would love to start playing "FPS games" but has a difficult time playing them.
- Has already tried solutions to help him, but none seem to work or stick with him.
- How does the client currently holds the controller and whether their grip changes over time due to fatigue.
- The exact range of motion the client has when attempting to use the left joystick and trigger.

<u>Unknown Information:</u>

- Preferred button sensitivity
- Preferred material texture feeling.

3.1.2 Problem Statement

Standard video game controllers are designed to be used with two hands, which makes them difficult to use for people who can only use one hand. Many games require the use of both joysticks, triggers, and bumper buttons at the same time, which limits what can be done with a single hand. This makes it hard to play a wide range of games and can prevent people from fully enjoying the gaming experience.

The client can only use his right hand and would like to play Nintendo Switch games independently using a controller he already owns. The goal of this project is to design an add-on attachment that enables one-handed users to access important controller functions without requiring assistance from others or purchasing a fully customized controller. The solution should be simple, adjustable, and reliable, while remaining affordable and environmentally friendly.

3.1.3 Need-Inspired Metrics & Benchmarking

- ***Need-Inspired Measurable Metrics:***

Based on the prioritized client needs identified earlier (specifically independent trigger access, comfort during long sessions, durability under repeated use, ease of installation, and affordability) the following measurable engineering metrics were developed. These metrics translate qualitative client needs into quantifiable performance criteria that can later be benchmarked and used to define target specifications.

Table 8. Measurable Metrics Derived from Client Needs

#	Client's Need	Metric	Unit
1.	Access all triggers independently	Maximum trigger actuation force through the attachment.	N (Newtons)
2.	Reach all controls	Effective trigger reach extension.	mm

3.	Comfortable long-term use	Material comfort and surface texture rating.	1–5 scale
4.	Easy installation	Installation time	sec
5.	Independent setup	Tools required	Y/N
6.	Affordability	Manufacturing cost	$

1. Trigger Actuation Force Through Attachment:

Unit: Newtons (N)

The primary functional requirement of the device is to allow the client to press the L2 and R2 triggers using one hand. Therefore, the force required to actuate the trigger through the mechanical attachment, the add-on, must be measured. This metric evaluates how much additional resistance the mechanism introduces compared to pressing the trigger directly. If the force is too high, it may cause strain, fatigue, or discomfort during extended gaming sessions. Since the client plays for multiple hours at a time, even small increases in required force could significantly affect usability. Measuring this in Newtons allows us to directly compare the mechanical efficiency of our design to standard controller trigger forces and to benchmark against other accessible solutions.

2. Effective Trigger Reach Extension:

Unit: millimeters (mm)

The client's primary difficulty is physically reaching the left trigger buttons. This metric measures how far the attachment extends the actuation point toward the user's natural grip position. The extension must be large enough to bridge the accessibility gap but not so large that it interferes with joystick movement or overall controller function. If the extension is too short, the device will not solve the problem. If it is too long, it may reduce precision or comfort. Measuring this distance

in millimeters ensures that the attachment directly addresses the physical reach limitation in a quantifiable way.

3. Material Comfort and Texture:

Unit: User rating (1–5 scale)

Since the attachment will be in direct contact with the user's hand for extended periods of time, the texture and surface finish of the material must be comfortable. Rough surfaces, sharp edges, or overly rigid textures may cause discomfort, especially during long gaming sessions. This metric evaluates how comfortable the material feels during use. It can be measured using a simple user feedback scale, where the client rates the comfort level from 1 (very uncomfortable) to 5 (very comfortable). Including this metric ensures that the design does not only function properly but also feels natural and comfortable, which is essential for long-term usability.

4. Installation Time:

Unit: seconds (sec)

Since independence is a high-priority client need, the device must be easy to install and remove without assistance. This metric measures the total time required for a user to properly attach and adjust the device to the controller. If installation takes too long or is complicated, the client may become discouraged from using it regularly. A short installation time supports accessibility, usability, and overall user satisfaction. Measuring installation time in seconds allows objective testing by timing multiple setup attempts and calculating an average value.

5. Tools Required:

Unit: Yes / No (Y/N)

The device should not require external tools such as screwdrivers, wrenches, or additional hardware for installation. Requiring tools would reduce user independence and make daily setup

more difficult. This metric evaluates whether the attachment can be installed and removed entirely by hand. A “No tools required” condition ensures accessibility and aligns with the Design for Assembly objective. Using a binary Yes/No metric provides a clear and measurable standard that directly reflects the client’s independence requirement.

6. Manufacturing Cost:

Unit: CAD ($)

The attachment must remain affordable and significantly less expensive than fully customized accessible controllers. This metric estimates the total material and production cost required to manufacture one unit. Keeping the manufacturing cost low ensures that the final product remains accessible to users and aligns with the Design for Cost objective identified earlier. Measuring cost in CAD provides a clear financial benchmark and allows comparison with existing commercial accessibility solutions.

- ***Benchmarking Existing Solutions:***

To determine whether they are realistic and achievable performance targets, existing accessibility gaming solutions were reviewed and compared. The purpose of benchmarking is to understand the current state of accessible controller design, identify strengths and weaknesses, and establish reasonable performance ranges for our measurable metrics such as trigger force, comfort, installation time, and durability.

The following solutions were analyzed because they represent the most relevant categories:

1. Fully redesigned adaptive controllers
2. 3D printed mechanical add-on attachments that already exist

1. Xbox Adaptive Controller (Microsoft):

The Xbox Adaptive Controller is an accessible commercial device that is marketed to help gamers with low mobility. It has huge programmable buttons and various external ports which enable users to attach switches and customize their system. The controller is professionally produced, and it is very durable and with low actuation force inputs to reduce strain. It is, however, much costlier than the common controllers and sometimes needs other accessories to operate at full capacity. For our project, this benchmark shows the importance of maintaining low trigger force and long-term durability while keeping the design affordable. It also highlights that our attachment-based approach is more cost-effective and compatible with the client's existing Nintendo Switch controller.

Figure 10: Xbox Adaptive Controller Interface Layout (Microsoft)

2. 3D Printed One-Handed Controller Attachment for Xbox:

Reference Video: "I made an one-handed Xbox controller" by Akaki Kuumeri

The reference video features a 3D printed one-handed controller attachment, which is a mechanical transfer of the trigger and button presses to the accessible part of the controller. This design is directly applicable in our project as it involves a similar form of attachment based on a

mechanical linkage system instead of overhauling the entire controller. It demonstrates that one-handed gaming is viable and achievable by extending the access to the trigger using printed lever arms. However, it also highlights important considerations such as potential increases in trigger force due to friction and the need for proper alignment to ensure smooth motion. This benchmark supports our design direction and emphasizes the importance of optimizing mechanical efficiency, comfort, and durability in our final attachment.

Figure 11: Picture of the Attachment from the Reference Video

From reviewing these solutions, we learned that one-handed gaming through mechanical attachments (add-ons) is both realistic and achievable. The examples demonstrate that mechanically connected trigger inputs are capable of being successful, however, they need to be designed with care to ensure smooth movement, minimize any extra resistance, and prevent untimely weariness in the long-term. Altogether, the benchmarking proves that our attachment-based method is rather feasible and cost-efficient, since it will increase accessibility and enable the client to retain his current controller rather than buy an expensive custom model.

3.1.4 Target Specifications

Table 9. Target Specifications for Add-On Attachment

#	Metric	Unit	Ideal Value	Marginal Value	Justification
1.	Maximum trigger actuation force through the attachment.	N (Newtons)	6.0 ± 0.5 N	≤ 6.5 N	The ideal keeps the trigger feeling the same as the original controller. The marginal limit makes sure the add-on doesn't feel too heavy or cause fatigue during long gaming sessions.
2.	Effective trigger reach extension.	mm	30 - 45mm	28 - 50mm	Must extend trigger into natural grip zone without interfering with joystick use. Too short leads to inaccessibility. Too long leads to awkward or blocks motion/buttons.
3.	Material comfort and surface texture rating.	1–5 scale	≥ 3.75/5	≥ 3/5	Client plays for long periods. Comfort must be high to prevent strain or irritation. A 3/5 is acceptable but not optimal.
4.	Installation time	sec	≤ 90 sec	≤ 180 sec	Must support independence. Ideal time to set up is 1.5 minutes. More than 3

					minutes becomes frustrating and reduces usability.
5.	Tools required	Y/N	No	No	Tools would reduce independence. This is a strict requirement so the marginal value must be the same as the ideal value.
6.	Manufacturing cost	$	$45 - $50	≤ $55	Ideal would be to remain below the cost of the project budget ($50). Exceeding $55 would conflict with our Design for Cost objective.

3.2 Concept Development

3.2.1 Functional Decomposition

To systematically develop concepts, the system was decomposed into functional subsystems based on our problem statement and for DFXs. The system was divided into four primary subsystems which include:

1. Trigger Actuation Transfer Mechanism:

The trigger actuation transfer mechanism is responsible for transmitting force from the accessible side of the controller to both bumpers and triggers (L1L2 and R1/R2). The selected concept uses rigid pivoted lever arms that rotate about fixed pivots and directly press each trigger when the user applies force to the corresponding extension paddles. The design preserves the natural feel of the original triggers while allowing independent one-handed operation. Each pivot

point is reinforced with a metal pin to reduce wear and ensure long-term durability during repeated daily gaming sessions.

2. Mounting / Attachment System:

The mounting system secures the attachment to the Nintendo Switch Pro Controller without requiring tools. The final concept is a snap-fit clip-on base that conforms to the curvature of the controller and locks into place through geometric retention. This design allows installation in under 60–90 seconds and supports independent setup by the user. It also ensures stability during long gaming sessions without blocking joysticks or button access.

3. Reach Extension System:

The reach extension subsystem ensures that the trigger actuation point aligns with the client's natural grip position. The selected concept is a fixed-length extension arm designed specifically to position the trigger within the ideal 30–45 mm reach range identified in the target specifications. By using a single optimized length rather than an adjustable mechanism, the design reduces part count, improves structural strength, and minimizes potential mechanical failure points. This fixed configuration increases reliability and lowers manufacturing cost while still meeting the client's ergonomic requirements.

4. Structural Support & Reinforcement:

The structural subsystem ensures the attachment maintains performance under repeated daily use. The main body is designed to be 3D printed in ABS (still is subject to change) for strength, flexibility, and low weight. High-stress areas, particularly the pivot point, are reinforced with a small metal pin to reduce fatigue and wear. This approach balances durability, sustainability, and cost while keeping the overall mass low.

This decomposition into sub-systems will ensure that each function can be evaluated independently against target specifications and DFX requirements.

The final assembled system integrates the snap-fit mounting base, fixed extension arms, pivoted lever mechanisms, and reinforced structures into a small attachment attached to the controller. When installed, the user presses the extension paddles, and this directs the lever arms to rotate and directly presses both triggers (L2 and R2). It is completely mechanical, lightweight and tool-free and can be set up and used independently. In general, the design preserves the entire trigger functionality and is still natural and ergonomic to use like the original controller.

3.2.2 Analysis and Evaluation of Concepts Against Target Specifications

Simple force estimates, dimensional checks, and feasibility analysis were used to assess the selected subsystem concepts in order to understand whether they satisfy the target specifications set by us, which are trigger force, reach extension, installation time, durability, and cost.

Trigger Actuation Force:

The Nintendo switch Pro Controller trigger has a full actuation force of around 5-6 N. No major mechanical losses should be anticipated because the lever arms are directly transmitting the force using a pivot where the friction is minimal and there is no cable system. Supposing that the losses in minor friction (≈ 5%) are taken into consideration, the input force would be needed:

$$(6.00N) \div (95.00\%) \approx 6.32N$$

This is in the marginal specification (≤ 6.5 N), and it is similar to the ideal (6.0 ± 0.5 N). Because the design does not introduce complex linkages or tension cables, it contributes to the comfort of the user in long-term play.

Reach Extension Verification:

The fixed extension arm was developed at the optimal 30-45 mm reach range that was set in the target specifications. As shown by physical measurement of the natural grip position of the client, a limitation of the natural grip position of about 37.5 mm causes the actuation paddle to be in a comfortable position that does not interfere with anything. A thing to take note of as well is that our group tested it as well. As this value is in the ideal range, the reach extension meets the ergonomic requirement without involving any means of adjustment.

Material Comfort Evaluation:

This will be evaluated through direct user testing to determine the material comfort prior to the final choice of materials. The 3D printing of small samples of PLA and ABS will be made and finished with a similar surface quality, which allows the client to feel and press both materials physically. The client will then evaluate the materials on a scale of 1–5 comfort scale based on texture, grip feel, and pressure. The main material used in the present prototype will be ABS as it is usually used in already existing accessories for controllers and has been tested by other users extensively on durability and comfort, while final selection will be confirmed after client feedback.

Installation Time and Tool Requirement:

The snap-fit mounting system will be tested qualitatively based on an assessment of the required number of steps to set up:

1. Align base to controller curvature
2. Press downward until snap engagement
3. Confirm secure fit

This is a toolless process and is approximately to require less than 60 to 90 seconds which is the optimal installation requirement.

Manufacturing Cost Estimation:

Based on current market prices observed on online stores such as Amazon, both ABS and PLA filament priced on average about $25 CAD per 1 kg spool. The material used is affordable as the unit will need a very small portion of the spool. Calculations will be made when we know the exact amount that will be used. With some extra hardware features like small metal pivot pin and light springs, and other minor details, it is estimated that the total cost of prototype will be under the exact of the ideal value range, which is great making it affordable.

3.2.3 Concept Selection for Further Development

The chosen solution that will be developed in the future, based on the estimation of all subsystem concepts against the defined target specifications, is the pivoted lever arm mechanism alongside the snap-fit mounting base and predetermined length extension arm. These ideas were selected since they satisfy all the trigger force, reach extension, installation time, durability, and cost requirements and is a complete but simple mechanical design. Other different ideas, like adjustable extension arms or cable systems, were not chosen because of more complicatedness, a stronger possibility of friction loss, and a decreased reliability of the product or service. In general, the selected design offers optimal accessibility and durability, as well as easy assembly and low costs.

3.2.4 Global Design Concept and Justification

The final global design concept is an integrated mechanical attachment that is embedded in the middle of the posterior of the Nintendo Switch Pro Controller. It integrates the snap-fit mounting base with the two pivoted lever arms, fixed length extension paddles (≈ 37.5mm), solid pin pivot, and small return springs into a compact assembly. As shown in the CAD models (pictures will be shown below), the base would fit around the shape of the controller and would place the lever arms in a balanced manner to perform an action on triggers (L2 and R2). As the user presses the extension paddles the lever arms rotate on fixed pivot and directly press triggers. The small springs guarantee that there is smooth rebounding movement and that the natural trigger feel is maintained.

This global concept is a refinement of the selected subsystem solutions, putting them together in an integrated and producible design. The fixed-length extension minimizes the number of parts and dismisses unnecessary complexity compared to an adjustable length extension. The fully mechanical configuration minimizes friction losses, potential failure points, and long-term wear while remaining lightweight and tool-free set up wise. The design is as well within the defined force, cost, and durability limits established in the target specifications. Overall, the combined system offers a perfectly balanced solution that focuses on accessibility, simplicity, reliability, and affordability.

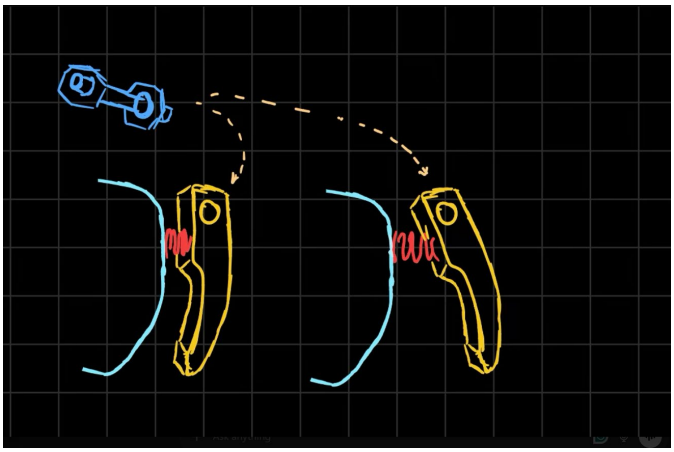


Figure 12: Conceptual Sketch of Dual Lever Arm Mechanism with Spring Return

Figure 13: Front View of the CAD Model

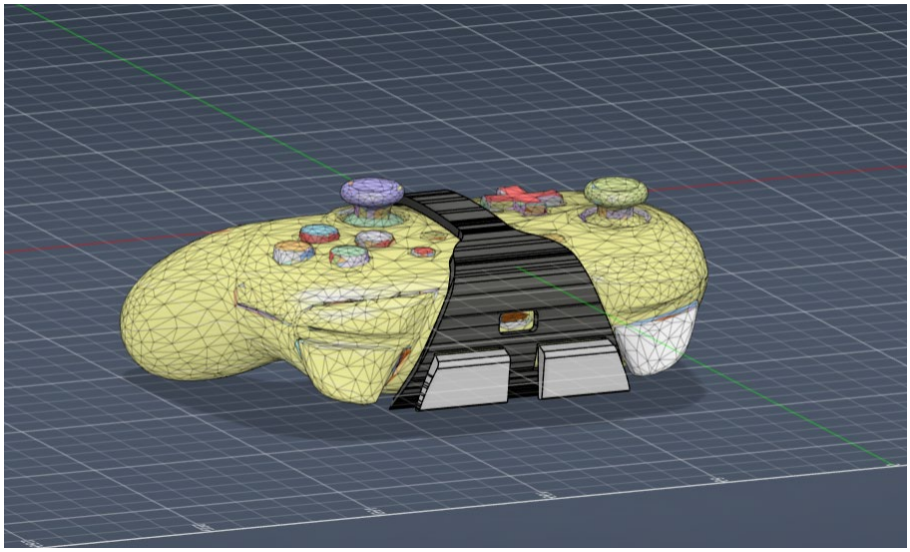


Figure 14: Angled View of the CAD Model

3.2.5 Relationship to Target Specifications and DFX

- **Relationship to Target Specifications, Benefits, and Drawbacks:**

The final global concept fits the target specifications formulated within the project earlier. The pivoted lever arm system keeps the force of a trigger actuation within the marginal limit (≤ 6.5N), and hence, the attachment cannot contribute significantly to user fatigue when playing video games over extended periods. The extension length that is fixed at about 37.5mm is within the optimal range of reach (30-45 mm) and the client can move the triggers without complicating the movement of joysticks. The snap-fit mounting system complies with the requirement of installation time (≤ 90sec) and meets the high feature of no tools required, which strengthens the independence of the user. Benefits of this concept includes how it is simple and mechanical reliability while being affordable to everyone making the device attainable for a broader user base. However, there are potential drawbacks to this concept such as the additions of the pivot pins and springs. Since the device would be used daily, repeated movement might slowly increase friction or cause some parts to loosen. This could slightly affect how smoothly the triggers feel after long-term use. Because of that, it might need occasional checking or small adjustments to keep it working properly.

- **Relationship to DFX Factors:**

The global concept strongly reflects the five primary Design for X factors identified earlier (accessibility, assembly, reliability and durability, sustainability, and cost). It prioritizes Design for Accessibility by restoring the independent trigger control without sacrificing the natural feel of the controller. The snap-fit has been used in Design for Assembly, allowing assembly to be quickly installed with no tools required as previously mentioned in this document. Using 3D printable ABS filament supports Design for Sustainability and Design for Reliability and Durability as it is a strong material that is also recyclable. The filaments are partially made from recycled plastics. This reduces the environmental impact compared to using other materials. Lastly, the simple mechanical design and low materials usage means that manufacturing cost will always be within the stipulated budget, which achieves the Design for Cost goal.

3.3 Project plan

This link will allow you to view our group's Notion:
[CRTL Sync (group 3.1): Notion Planning for Task, Deliverables, etc.]

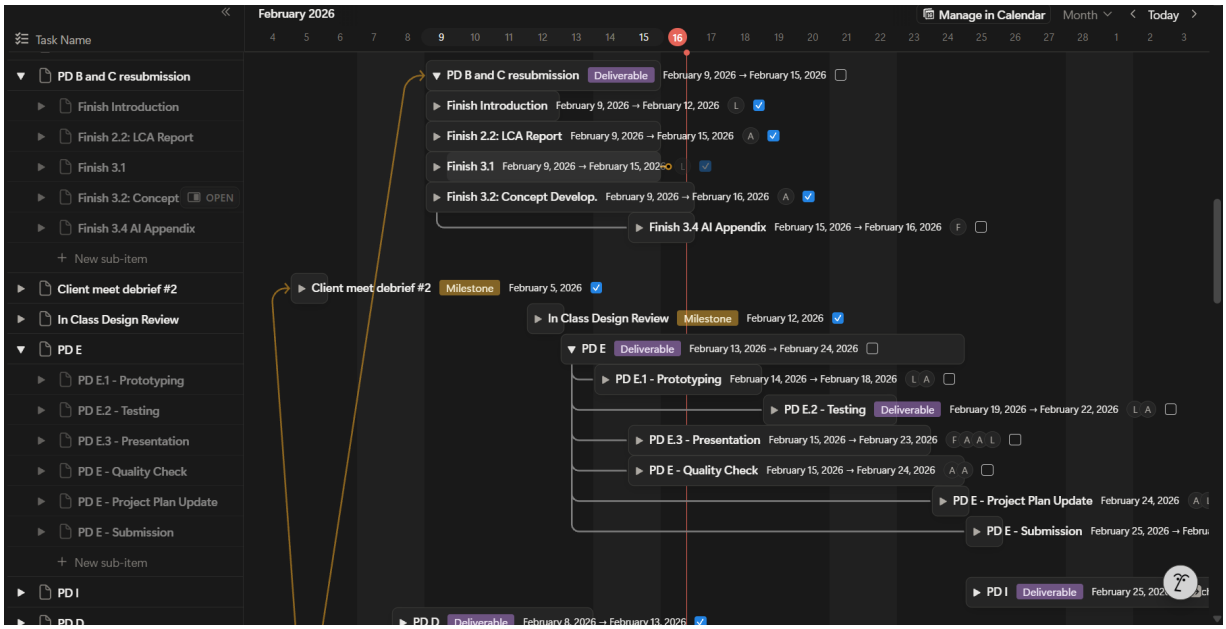


Figure 15: Updated Gantt Chart

Figure 16: Project Task List Showing Deliverables, Milestones, and Deadlines

3.4 AI Appendix

AI was used in this section of the document to primarily enhance clarity, structure, and form or format of the report. It assisted in the creation of title of figures, tables and section headings to ensure consistency. The approximate values of trigger actuation force and overall material properties were also investigated with the help of AI and were used to make early estimates during the design. In subsystem development, AI was occasionally used to give us advantages and disadvantages of the various mechanical solutions, which allowed us to brainstorm about potential solutions more effectively.

Nevertheless, AI did not take the final design decisions. Although it served to discover pros and cons of some of the ideas that we would never have considered, the group effectively criticized those ideas and chose subsystems according to our engineering research and the calculation and demands of the clients. The team had to done all the target specifications, feasibility analysis, and final concept development. AI only assisted us in identifying appropriate units for that part. In summary, the technical rationale and decision-making were on our part, and AI was again utilized as a supporting brainstorming and editing program.

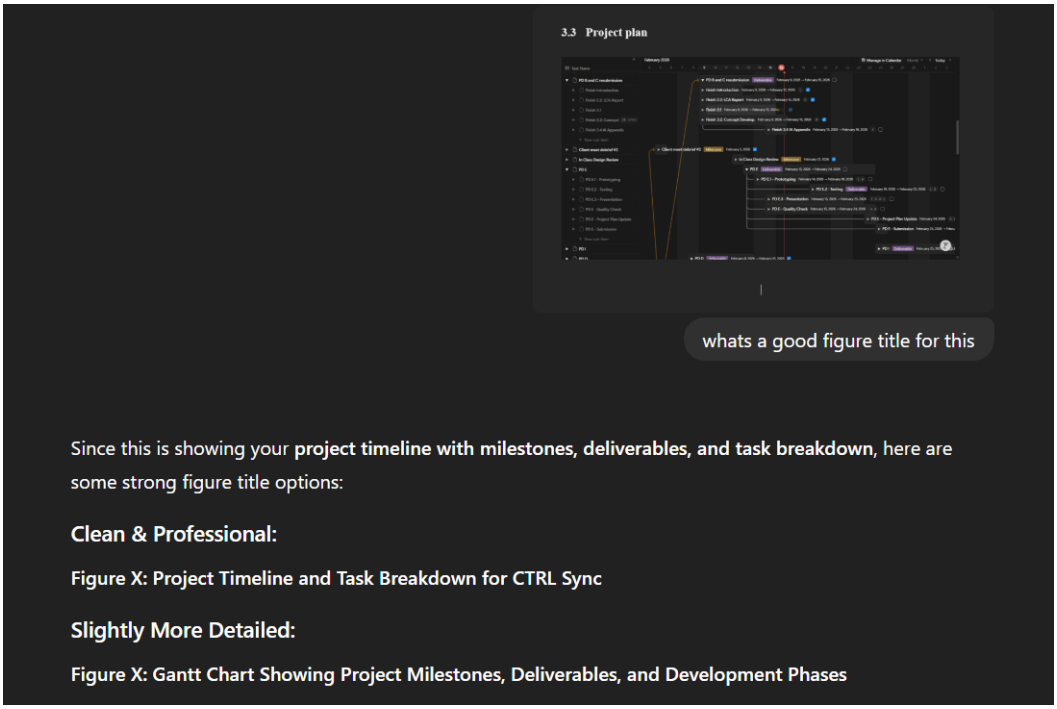


Figure 17: Example of AI-Assisted Figure Title Generation

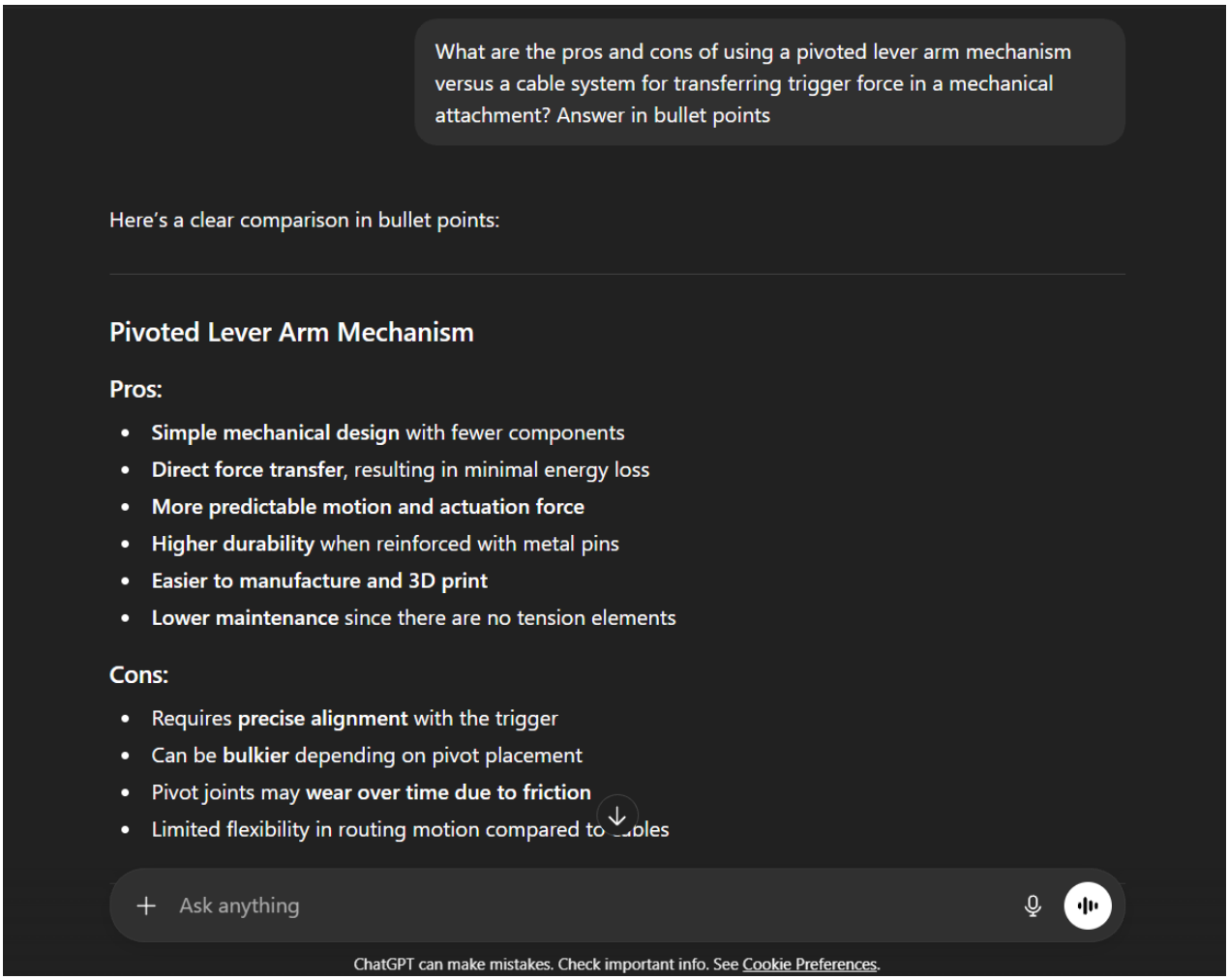


Figure 18: Example of AI Use for Pros and Cons Analysis During Concept Selection

4 Detailed Design and BOM

4.1 Client Feedback and Design Changes

After meeting with the client and seeing him use the controller in person, the primary design focus shifted based on his feedback. The main difficulty he identified was not with joystick movement, as initially assumed, but with physically reaching the trigger buttons (L2 and R2), especially during longer gameplay sessions. As a result, the updated concept now prioritizes a modular attachment system specifically designed to improve access to these trigger inputs.

Originally, the team believed that joystick control and in-game movement would be the main accessibility challenge. However, our direct interaction clarified that his current movement control is already comfortable and does not require modification. The client emphasized that he does not want a device that changes or assists with in-game mechanics, but rather a solution that improves physical reach and usability while maintaining the natural feel of the controller. He also mentioned that maintaining comfort over extended periods is important, since he often plays for several hours at a time.

This feedback refined the project's direction by shifting our focus toward improving trigger accessibility without interfering with joystick control. It ensured that the design remains user-centered and addresses the client's actual needs rather than our initial assumptions.

4.2 Detailed Design

4.3 Manufacturing Considerations

4.4 Skills and Resources Available

The main technical resource required for this project is access to 3D modelling and 3D printing software, since the final design focuses on a mechanical attachment rather than an electronic system. As a team, we already have experience using CAD software and 3D printing tools, which will allow us to efficiently design, test, and modify components as we refine the modular attachment system. This experience is especially important because the design will likely go through multiple iterations to ensure proper fit, range of motion, and comfort for the client.

Although our group has strong circuit design and coding experience, these skills will not be heavily used in this project since the current concept is primarily mechanical. Instead, the focus will be on precise mechanical modelling, joint design, and ensuring that the attachment can withstand repeated movement without failure. Understanding tolerances, part alignment, and structural strength will be key considerations during development.

Access to 3D printers through the university makerspace and lab facilities will be an important resource for testing different configurations and improving accessibility for the client. Rapid prototyping will allow us to quickly evaluate fit and functionality directly on the controller and make necessary adjustments without significant delays. Additionally, access to basic workshop tools for sanding, fitting, and minor adjustments will help improve the overall finish and performance of the prototype.

Currently, no major technical skills are missing. However, if advanced modelling features, tolerance adjustments, or printing support optimization become necessary, we plan to use online

tutorials, workshops held by the university and TA guidance to improve our designs and ensure the final prototype is both functional and well-constructed.

4.5 Time and Feasibility Assessment

4.6 BOM

BOM_W26.xlsx

5 Conclusions

Summarize your lessons learned and your work related to your project. Discuss any outstanding issues or implications for the project.

6 Bibliography

[1] https://explodingtopics.com/blog/number-of-gamers

[2] https://dimensiva.com/3dmodels/nintendo-switch-pro-controller-by-nintendo/